

HAND WRITTEN MATERIALS

MA4201 PROBABILITY AND STATISTICS

COURSE OBJECTIVES

The main objectives of this course are to:

1. To introduce the basic concepts of probability and distributions.
2. To learn the basic concepts of two -dimensional random variables.
3. To acquire the knowledge in random processes, stationary, Markov and poisson process.
4. To acquire the knowledge of testing of hypothesis for small and large samples this plays an important role in real life problems.
5. To summarise the design of experiments in the field of agriculture.

COURSE OUTCOMES

Upon completion of the course, the students will be able to:

CO1: Understand the fundamental knowledge of the concepts of probability and have knowledge of Standard distributions which can describe real life phenomenon.

CO2: Recognize the basic concepts of two - dimensional random variables and apply in engineering applications.

CO3: Develop the basic concepts of random processes which are widely used in engineering fields.

CO4: Apply the concept of testing of hypothesis for small and large samples in real life problems.

CO5: Investigate of design of experiments in the field of agriculture.

UNIT V DESIGN OF EXPERIMENTS

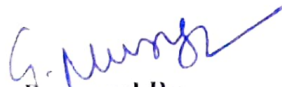
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One way and two-way classifications - Completely randomized design - Randomized block design - Latin square design.

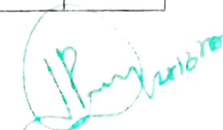
CO-PO Mapping

COs	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
C201.1	3	2	1									
C201.2	3	2	1									
C201.3	3	2	1									
C201.4	3	2	1									
C201.5	3	2	1									
Avg	3	2	1									

1-Low,2-Medium,3-High


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Unit - 5 Design of Experiments

One Way Classification:

Completely Randomized Design (CRD)

H_0 : There is no significant difference b/w treatments

H_1 : There is a significant difference b/w treatments

Working Rule:

Step: 1 To Find N ; N - Number of observations

Step: 2 Find T ; T - Total of all observation

Step: 3 Find $\frac{T^2}{N}$, $\frac{T^2}{N}$ - Correction factor

Step: 4 Find TSS ; Total sum of squares

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \dots - \frac{T^2}{N}$$

$$TSS = \sum Y_1^2 + \sum Y_2^2 + \sum Y_3^2 + \dots - \frac{T^2}{N}$$

Step: 5 Find SSC ; Sum of squares of column

$$SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \dots - \frac{T^2}{N}$$

where N_1 - No. of elements in each column. (or)

Find SSR ; Sum of squares of Row

$$SSR = \frac{(\sum Y_1)^2}{N_1} + \frac{(\sum Y_2)^2}{N_1} + \dots - \frac{T^2}{N}$$

where N_1 - No. of elements in each Row.

Step : 6 Find SSE ; Error Sum of squares

$$SSE = TSS - SSC$$

(or)

$$SSE = TSS - SSR$$

Step: 7 ANOVA Table

Step: 8 Conclusion

i) Tab Value $F_c >$ Calculated Value F_c

We accept H_0

ii) Tab. Value $<$ Calculated val

We Reject H_0 .

CRD : ANOVA TABLE

SOURCE OF VARIANCE	SUM OF SQUARES	DEGREES OF FREEDOM	MEAN SUM OF SQUARES	VARIANCE RATIO	TABLE VALUE @ 1% OR 5%
BETWEEN COLUMN (OR) ROW	SSC (or) SSR	C-1 (or) R-1	MSC = $\frac{SSC}{C-1}$ or MSR = $\frac{SSR}{R-1}$	$F_c = \frac{MSC}{MSE}$ (or) $F_c = \frac{MSE}{MSC}$	$F_c = (r_1, r_2)$ or
ERROR	SSE	N-C	MSE = $\frac{SSE}{N-C}$	$F_R = \frac{MSR}{MSE}$ (or) $F_R = \frac{MSE}{MSR}$ $\therefore F > 1$	$F_R = (r_1, r_2)$

Problems:

1. The following are the no. of mistakes made in 5 successive days by 4 Technicians working for a photographic Lab. Test at a level of significance $\alpha = 0.01$. Test whether the difference among 4 sample means can be attributed

to change

Technician

<u>I</u>	<u>II</u>	<u>III</u>	<u>IV</u>
6	14	10	9
14	9	12	12
10	12	7	8
8	10	15	10
11	14	11	11

Soln: H_0 : There is no significant difference b/w treatments. H_1 : There is a significant difference b/w treatments.

Subtracting with 10

X_1	X_2	X_3	X_4	TOTAL	X_1^2	X_2^2	X_3^2	X_4^2
-4	4	0	-1	-1	16	16	0	1
4	-1	2	2	7	16	1	4	4
0	2	-3	-2	-3	0	4	9	4
-2	0	5	0	3	4	0	25	0
1	4	1	1	7	1	16	1	1
$T = -1$	9	5	0	(13)	37	37	39	10
ΣX_1	ΣX_2	ΣX_3	ΣX_4		ΣX_1^2	ΣX_2^2	ΣX_3^2	ΣX_4^2

Step: 1 $N = 20$ Step: 2 $T = 13$ Step: 3 $\frac{T^2}{N} = \frac{13^2}{20} = \frac{169}{20} = 8.45$

Step: 4 :- TSS

$$TSS = \sum x_1^2 + \sum x_2^2 + \sum x_3^2 + \sum x_4^2 - \frac{T^2}{N}$$

$$= 37 + 37 + 39 + 10 - 8.45$$

$$TSS = 114.55$$

Step: 5 :- SSC

$$SSC = \frac{(\sum x_1)^2}{N_1} + \frac{(\sum x_2)^2}{N_1} + \frac{(\sum x_3)^2}{N_1} + \frac{(\sum x_4)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{(-1)^2}{5} + \frac{9^2}{5} + \frac{5^2}{5} + 0 - 8.45$$

$$SSC = 12.95$$

Step: 6 :- SSE

$$SSE = TSS - SSC$$

$$= 114.55 - 12.95$$

$$SSE = 101.6$$

Step: 7 ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value at 1% level
B/w column	SSC = 12.95	C-1 = 4-1 = 3	MSC = $\frac{SSC}{C-1}$ = $\frac{12.95}{3}$ = 4.32	$F_c = \frac{MSE}{MSC}$ = $\frac{6.35}{4.32}$	$F_c = (\gamma_1, \gamma_2)$ = (16, 3)
Error	SSE = 101.6	N-C = 20-4 = 16 (γ_1)	MSE = $\frac{SSE}{N-C}$ = 101.6/16 = 6.35	$F_c = 1.471$	$F_c = 26.78$

Step: 8

Calculated Value = 1.471

Table Value = 26.78

$$T.V > C.V$$

$\therefore H_0$ may be accepted.

2. The following table shows life in hours of 4 brands of electric lamp.

Brand

A	1610	1610	1650	1680	1700	1720	1800
B	1580	1640	1700	1750			
C	1460	1550	1600	1620	1640	1660	1740
D	1510	1520	1530	1570	1600	1680	

Perform an analysis of variance test for the mean life of the 4 brand lamps.

Soln:

H_0 : There is no significant diff. b/w the brands

H_1 : There is a significant diff. b/w the brands

sub 1600 and divide by 10.

A X_1	B X_2	C X_3	D X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
1	-2	-14	-9	-24	1	4	196	81
1	4	-5	-8	-8	1	16	25	64
5	10	0	-7	8	25	100	0	49
8	15	2	-3	22	64	225	4	9
10		4	0	14	100		16	0
12		6	8	26	144		36	64
20		14		34	400		196	
		22		22			484	
T = 57	27	29	-9	(94)	735	345	957	267

Step: 1 :-

$$N = \text{No. of terms} = 25$$

Step: 2 :-

$$T = 94$$

$$\text{Step: 3 :- } \frac{T^2}{N} = \frac{(94)^2}{25} = 353.44$$

Step: 4 :- TSS

$$\begin{aligned} \text{TSS} &= \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N} \\ &= 735 + 345 + 957 + 267 - 353.44 \end{aligned}$$

$$\boxed{\text{TSS} = 1950.56}$$

Step: 5 :- SSC

$$\begin{aligned} \text{SSC} &= \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N} \\ &= \frac{(57)^2}{7} + \frac{(27)^2}{4} + \frac{(29)^2}{8} + \frac{(-19)^2}{6} - 353.44 \end{aligned}$$

$$\boxed{\text{SSC} = 458.25}$$

Step: 6 : SSE

$$\text{SSE} = \text{TSS} - \text{SSC}$$

$$= 1950.56 - 458.25$$

$$\boxed{\text{SSE} = 1492.31}$$

Step: 7 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table value at 5%
B/w column	SSC = 458.25	C-1 = 4-1 = 3	MSC = $\frac{SSC}{C-1}$ = $\frac{458.25}{3}$ = 152.75	$F_c = \frac{MSC}{MSE}$ = $\frac{152.75}{71.062}$ = 2.14	$F_c = (2, 21)$ = (3, 21) = 3.07
Error	SSE = 1492.31	N-C = 25-4 = 21	MSE = $\frac{SSE}{N-C}$ = $\frac{1492.31}{21}$ = 71.062		

Step: 8 :- Conclusion

Calculated value = 2.14

Table Value = 3.07

$\therefore$ Table Value > Calculated Value

$\therefore H_0$ may be accepted.

3.

Given

		Engine		
		1	2	3
Detergent	A	45	43	51
	B	47	46	52
	C	48	50	55
	D	42	37	49

Perform ANOVA and test at 5% level of significance whether these differences in the detergents or in the engine.

Soln:

H_0 : There is no significant difference b/w the detergents or in the engine.

H_1 : There is a significant difference b/w the detergents or in the engine.

Sub. with 40.

X_1	X_2	X_3	Total	X_1^2	X_2^2	X_3^2
5	3	11	19	25	9	121
7	6	12	25	49	36	144
8	10	15	33	64	100	225
2	-3	9	8	4	9	81
22	16	47	(85)	142	154	571

Step: 1 :- $N = 12$ Step: 2 :- $T = 85$ Step: 3 :- $\frac{T^2}{N} = \frac{(85)^2}{12} = 602.08$

Step: 4 :- TSS

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 - \frac{T^2}{N}$$

$$= 142 + 154 + 571 - 602.08$$

$$TSS = 264.92$$

Step: 5 :- SSC

$$= \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{(22)^2}{4} + \frac{16^2}{4} + \frac{47^2}{4} - 602.08$$

$$SSC = 135.17$$

Step: 6 :- SSE

$$SSE = TSS - SSC$$

$$= 264.92 - 135.17$$

$$SSE = 129.75$$

Step: 7 :-

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value @ 5%
B/w Column	SSC = 135.17	$C-1$ = $3-1$ = 2	$MSC = \frac{SSC}{C-1}$ = $\frac{135.17}{2}$ = 67.58	$F_c = \frac{MSC}{MSE}$ = $\frac{67.58}{14.41}$	$F_c = (2, 9)$ $F_c(2, 9)$ $F_c = 4.26$
Error	SSE = 129.75	$N-C$ = $12-3$ = 9	$MSE = \frac{SSE}{N-C}$ = $\frac{129.75}{9}$ = 14.41	$F_c = 4.69$	

Step: 8 :-

Calculated value = 4.69

Tabulated value = 4.26

Table value < Calculated value.

 $\therefore H_0$ is Rejected.

4. The accompanying data resulted from an experiment comparing the degree of boiling for fabric copolymerized with the 3 different mixtures of methacrylic acid. Analyse the classification.

Mixture 1 : 0.56 1.12 0.90 1.07 0.94

2 : 0.72 0.69 0.87 0.78 0.91

3 : 0.62 1.08 1.07 0.99 0.93

Soln: H_0 : There is no significant difference b/w the mixtures.

H_1 : There is a significant difference b/w the mixtures.

[Subtracting with 0.8 and Multiplying by 100]

X_1	X_2	X_3	X_4	X_5	Total	X_1^2	X_2^2	X_3^2	X_4^2	X_5^2
-24	32	10	27	14	59	576	1024	100	729	196
-8	-11	7	-2	11	-3	64	121	49	4	121
-18	28	27	19	13	69	324	784	729	361	169
-50	49	44	44	38	125	964	1929	878	1094	486

Step: 1 :- $N = 15$

Step: 2 :- $T = 125$

Step: 3 :- $\frac{T^2}{N} = \frac{(125)^2}{15} = 1041.67$

Step: 4 :- TSS

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 + \sum X_5^2 - \frac{T^2}{N}$$

$$= 964 + 1929 + 878 + 1094 + 486 - 1041.67$$

$$TSS = 4309.33$$

Step: 5 :- SSC

$$SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} + \frac{(\sum X_5)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{(-50)^2}{3} + \frac{(49)^2}{3} + \frac{(44)^2}{3} + \frac{(44)^2}{3} + \frac{(38)^2}{3} - 1041.67$$

$$SSC = 2363.99$$

Step: 6 :- SSE

$$SSE = TSS - SSC$$

$$= 4309.33 - 2363.99$$

$$SSE = 1945.34$$

Step: 7 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value @ 5%
B/w Column	SSC = 2363.99	C-1 = 5-1 = 4	MSC = $\frac{SSC}{C-1}$ = $\frac{2363.99}{4}$ = 590.99	$F_c = \frac{MSC}{MSE}$ = $\frac{590.99}{194.534}$ $F_c = 3.037$	$F_c(\alpha_1, \alpha_2)$ = (4, 10) $F_c = 3.48$
Error	SSE = 1945.34	N-C = 15-5 = 10	MSE = $\frac{SSE}{N-C}$ = $\frac{1945.34}{10}$ = 194.534	$F_c > 1$	

Step: 8 :- Conclusion:

Calculated Value : 3.037

Table Value : 3.48

$$T.V > C.V$$

$\Rightarrow H_0$ may be Accepted.

TWO WAY CLASSIFICATION: (RBD)

RANDOMIZED BLOCK DESIGN.

Working Rule:

Step: 1 Find N ; N - No. of observations.

Step: 2 Find T ; T - Total of all observations.

Step: 3 $\frac{T^2}{N}$ [Correction Factor]

Step: 4 TSS

$$TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 \dots - \frac{T^2}{N}$$

Step: 5 SSC

$$SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \dots - \frac{T^2}{N} \quad [N_1 - \text{no. of element in each column}]$$

Step: 6 SSR

$$SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \dots - \frac{T^2}{N} \quad [N_2 - \text{no. of elements in each row}]$$

Step: 7 SSE

$$SSE = TSS - SSC - SSR$$

Step: 8 ANOVA TABLE

SOURCE OF VARIANCE	SUM OF SQUARES	DEGREE OF FREEDOM	MEAN SUM OF SQUARES	VARIANCE RATIO	TABLE VALUE @ 1%, 5%
B/w column	SSC	$C-1$	$MSC = \frac{SSC}{C-1}$	$F_C = \frac{MSC}{MSE}$ (or) $F_C = \frac{MSE}{MSC}$	$F_C (\gamma_1, \gamma_2)$
B/w Row	SSR	$R-1$	$MSR = \frac{SSR}{R-1}$	$F_R = \frac{MSR}{MSE}$ (or) $F_R = \frac{MSE}{MSR}$	$F_R (\gamma_1, \gamma_2)$
Error	SSE	$(C-1)(r-1)$	$MSE = \frac{SSE}{(C-1)(r-1)}$	$\therefore F > 1$	

Step: 9 :- Conclusion

i) $\text{Tab. } F_c > \text{cal. } F_c$

$\text{Tab. } F_R > \text{cal. } F_R$

H_0 may be accepted

ii) $\text{Tab. } F_c < \text{cal. } F_c$

$\text{Tab. } F_R < \text{cal. } F_R$

H_0 is Rejected.

1. Analyse the following RBD and find the conclusion.

Treatment

Blocks	T_1	T_2	T_3	T_4
B_1	10	12	18	20
B_2	15	25	17	13
B_3	13	12	15	10
B_4	16	14	20	10
B_5	17	13	18	12

Soln:

H_0 : There is no significant difference b/w blocks & treatments.

H_1 : There is a significant difference b/w blocks & treatments.

(X-10)

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	0	2	8	10	20	0	4	64	100
Y_2	5	15	7	3	30	25	225	49	9
Y_3	3	2	5	0	10	9	4	25	0
Y_4	6	4	10	0	20	36	16	100	0
Y_5	7	3	8	2	20	49	9	64	4
	21	26	38	15	100	119	258	302	113

Step: 1 :- $N = 20$

Step: 2 :- $T = 100$

Step: 3 :- $\frac{T^2}{N} = \frac{(100)^2}{20} = 500$

Step: 4 :- TSS

$$= \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$$

$$= 119 + 258 + 302 + 113 - 500$$

TSS = 292

Step: 5 :- SSC = $\frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$

$$= \frac{(21)^2}{5} + \frac{(26)^2}{5} + \frac{(38)^2}{5} + \frac{(15)^2}{5} - 500 = \boxed{57.12 = SSC}$$

Step: 6 :- SSR = $\frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} + \frac{(\sum Y_5)^2}{N_2} - \frac{T^2}{N}$

$$= \frac{(20)^2}{4} + \frac{(30)^2}{4} + \frac{(10)^2}{4} + \frac{(20)^2}{4} + \frac{(20)^2}{4} - 500$$

SSR = 50

Step: - 7 :- SSE = TSS - SSC - SSR = 292 - 57.12 - 50

SSE = 184.8

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table value @ 5%
B/w column	SSC = 57.12	$C-1 = 4-1 = 3$	$MSC = \frac{SSC}{C-1} = \frac{57.12}{3} = 19.06$	$F_C = \frac{MSC}{MSE} = 1.237$	$F_C(3, 12) = 3.49$
B/w Rows	SSR = 50	$r-1 = 5-1 = 4$	$MSR = \frac{SSR}{r-1} = \frac{50}{4} = 12.5$	$F_R = \frac{MSE}{MSR} = 1.232$	$F_R(12, 4) = 5.91$
Error	SSE = 184.8	$(C-1)(r-1) = 3 \times 4 = 12$	$MSE = \frac{SSE}{(C-1)(r-1)} = \frac{184.8}{12} = 15.4$	$F > 1$	

Step: 9 :- Conclusion

	Between Column	B/w Rows
T.V	3.49	5.91
C.V	1.2381	1.2320
Result	$T.V > C.V$	$T.V > C.V$
Conclusion	H_0 may be accepted	H_0 may be Accepted

2. The following data represents no. of units of production Per day turn out by 5 different workers using 4 different types of machines

		Machine type			
		A	B	C	D
Workers	1	44	38	47	36
	2	46	40	52	43
	3	34	36	44	32
	4	43	38	46	33
	5	38	42	49	39

Soln:

H_0 : There is no significant difference b/w machines

H_1 : There is a significant difference b/w machines

(X-40)

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	4	-2	7	-4	5	16	4	49	16
Y_2	6	0	12	3	21	36	0	144	9
Y_3	-6	-4	4	-8	-14	36	16	16	64
Y_4	3	-2	6	-7	0	9	4	36	49
Y_5	-2	2	9	-1	8	4	4	81	1
	$\sum X_1^5$	$\sum X_2^{-6}$	$\sum X_3^{38}$	$\sum X_4^{-17}$	(20)	$\sum X_1^{101}$	$\sum X_2^{28}$	$\sum X_3^{326}$	$\sum X_4^{139}$

Step: 1 :- $N = 20$

Step: 2 :- $T = 20$

Step: 3 :- $\frac{T^2}{N} = \frac{(20)^2}{20} = 20$

Step: 4 :- TSS

$$= \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$$

$$= 101 + 28 + 326 + 139 - 20$$

$$\boxed{TSS = 574}$$

Step: 5 :- SSC

$$= \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{25}{5} + \frac{36}{5} + \frac{(38)^2}{5} + \frac{289}{5} - 20$$

$$\boxed{SSC = 338.8}$$

Step: 6 :- SSR

$$= \frac{(\sum y_1)^2}{N_2} + \frac{(\sum y_2)^2}{N_2} + \frac{(\sum y_3)^2}{N_2} + \frac{(\sum y_4)^2}{N_2} + \frac{(\sum y_5)^2}{N_2} - \frac{T^2}{N}$$

$$= \frac{(5)^2}{4} + \frac{(21)^2}{4} + \frac{(-14)^2}{4} + 0 + \frac{(8)^2}{4} - 20$$

$$\boxed{SSR = 161.5}$$

Step: 7 :- SSE

$$= TSS - SSC - SSR$$

$$= 574 - 338.8 - 161.5$$

$$\boxed{SSE = 73.7}$$

Step: 8 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value @ 5%
B/w Columns	SSC = 338.8	C-1 = 4-1 = 3	MSC = $\frac{SSC}{C-1}$ = $\frac{338.8}{3}$ = 112.933	F _C = $\frac{MSC}{MSE}$ = $\frac{112.933}{6.141}$ F _C = 18.39	F _C (3, 12) = F _C (3, 12) = 3.49
B/w Rows	SSR = 161.5	r-1 = 5-1 = 4	MSR = $\frac{SSR}{r-1}$ = $\frac{161.5}{4}$ = 40.375	F _R = $\frac{MSR}{MSE}$ = $\frac{40.375}{6.141}$ F _R = 6.5747	F _R (4, 12) = F _R (4, 12) = 3.26
Error	SSE = 73.7	(C-1)(r-1) = 12	MSE = $\frac{SSE}{(C-1)(r-1)}$ = $\frac{73.7}{12}$ = 6.141	∴ F > 1	

Step: 9 :- Conclusion.

	B/w columns	B/w Rows
T.V	3.49	3.26
C.V	18.39	6.5747
Result	$T.V < C.V$	$T.V < C.V$
Conclusion	H_0 is Rejected	H_0 is Rejected.

3. Three varieties of coal were analysed by 4 chemists and the ash content is tabulated here. Perform an analysis of Variance.

		Chemist			
		A	B	C	D
Coal	I	8	5	5	7
	II	7	6	4	4
	III	3	6	5	4

Soln: H_0 : There is no significant difference b/w treatments H_1 : There is a significant difference b/w treatments

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	8	5	5	7	25	64	25	25	49
Y_2	7	6	4	4	21	49	36	16	16
Y_3	3	6	5	4	18	9	36	25	16
	18	17	14	15	(64)	122	97	66	81
	ΣX_1	ΣX_2	ΣX_3	ΣX_4		ΣX_1^2	ΣX_2^2	ΣX_3^2	ΣX_4^2

Step: 1 :- $N = 12$

Step: 2 :- $T = 64$

Step: 3 :- $\frac{T^2}{N} = \frac{(64)^2}{12} = 341.33$

Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 122 + 97 + 66 + 81 - 341.33$

$TSS = 24.66$

Step: 5 :- $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$
 $= \frac{(18)^2}{3} + \frac{(17)^2}{3} + \frac{(14)^2}{3} + \frac{(15)^2}{3} - 341.33$

$SSC = 3.3334$

Step: 6 :- $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} - \frac{T^2}{N}$
 $= \frac{(25)^2}{4} + \frac{(21)^2}{4} + \frac{(18)^2}{4} - 341.33$

$SSR = 6.1667$

Step: 7 :- SSE

$SSE = TSS - SSC - SSR$
 $= 24.66 - 3.3334 - 6.1667$

$SSE = 15.1666$

Step : 8 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value @ 5%
B/w Column	SSC = 3.33	C-1 = 4-1 = 3	MSC = $\frac{SSC}{C-1}$ = $\frac{3.33}{3}$ = 1.111	$F_C = \frac{MSE}{MSC}$ = $\frac{2.5278}{1.111}$ $F_C = 2.2750$	$F_C(\gamma_1, \gamma_2)$ = $F_C(6, 3)$ = 8.94
B/w Rows	SSR = 6.166	$\gamma-1$ = 3-1 = 2	MSR = $\frac{SSR}{\gamma-1}$ = $\frac{6.166}{2}$ = 3.0834	$F_R = \frac{MSR}{MSE}$ = $\frac{3.0834}{2.5278}$ = 1.2198	$F_R(\gamma_1, \gamma_2)$ = $F_R(2, 6)$ = 5.14
Error	SSE = 15.166	$(C-1)(\gamma-1)$ = 3 × 2 = 6	MSE = $\frac{SSE}{(C-1)(\gamma-1)}$ = 15.166/6 = 2.5278	$\therefore F > 1$	

Step : 9 :- Conclusion

	B/w Column	B/w Rows
T.V	8.94	5.14
C.V	2.2750	1.2198
Result	$T.V > C.V$	$T.V > C.V$
Conclusion	H_0 may be Accepted	H_0 may be Accepted

4. The Result of RBD experiment on 3 blocks with 4 treatment A, B, C, D are tabulated. Perform an analysis of variance for the following.

Treatment effects

I	A 36	D 35	C 21	B 36
II	D 32	B 29	A 28	C 31
III	B 28	C 29	D 29	A 26

Soln:

On Rearranging the treatment

	A	B	C	D
I	36	36	21	35
II	28	29	31	32
III	26	28	29	29

H_0 : There is no significant difference b/w treatments

H_1 : There is a significant difference b/w treatments

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	6	6	-9	5	8	36	36	81	25
Y_2	-2	-1	1	1	-1	4	1	1	1
Y_3	-4	-2	-1	-1	-8	16	4	1	1
	0	3	-9	5	(-1)	56	41	83	27

Step: 1 :- $N = 12$

Step: 2 :- $T = -1$

Step: 3 :- $\frac{T^2}{N} = \frac{(-1)^2}{12} = 0.0833$

Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 56 + 41 + 83 + 27 - 0.0833$

$TSS = 206.9167$

Step: 5 :- SSC

$$= \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$$

$$= 0 + \frac{9}{3} + \frac{81}{3} + \frac{25}{3} - 0.0833$$

$$\boxed{SSC = 38.2500}$$

Step: 6 :- SSR

$$= \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} - \frac{T^2}{N}$$

$$= \frac{64}{4} + \frac{1}{4} + \frac{64}{4} - 0.0833$$

$$\boxed{SSR = 32.1667}$$

Step: 7 :- SSE

$$= TSS - SSC - SSR$$

$$= 206.9167 - 38.2500 - 32.1667$$

$$\boxed{SSE = 136.5}$$

Step: 9 :- Conclusion

	B/w Columns	B/w Rows
T.V	8.94	19.33
C.V	1.78	1.414
Result	$T.V > C.V$	$T.V > C.V$
Conclusion	H_0 may be Accepted	H_0 may be Accepted

Step: 8 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value @ 5%
B/w Columns	SSC = 38.25	C-1 = 4-1 = 3	MSC = $\frac{SSC}{C-1}$ = $\frac{38.2500}{3}$ = 12.75	$F_C = \frac{MSE}{MSC}$ = $\frac{22.75}{12.75}$ = 1.78	$F_C(\gamma_1, \gamma_2)$ = $F_C(6, 3)$ = 8.94
B/w Rows	SSR = 32.166	$\gamma-1$ = 3-1 = 2	MSR = $\frac{SSR}{\gamma-1}$ = $\frac{32.1667}{2}$ = 16.08	$F_R = \frac{MSE}{MSR}$ = $\frac{22.75}{16.08}$ = 1.414	$F_R(\gamma_1, \gamma_2)$ = $F_R(6, 2)$ = 19.33
Error	SSE = 136.5	$(C-1)(\gamma-1)$ = 6	MSE = $\frac{SSE}{6}$ = $\frac{136.5}{6}$ = 22.75	$F_C > 1$ $F_R > 1$	

5. A Company appoints 4 Salesman A, B, C, D & observes their sales in 3 seasons Summer, winter and monsoon are given in following table.

	Salesman			
	A	B	C	D
Summer	45	40	38	37
Winter	43	41	45	38
Monsoon	39	39	41	41

Soln:

H_0 : There is no significant difference b/w seasons

H_1 : There is a significant difference b/w seasons

Step: 1 :- $N = 12$

Step: 2 :- $T = 7$

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	5	0	-2	-3	0	25	0	4	9
Y_2	3	1	5	-2	7	9	1	25	4
Y_3	-1	-1	1	1	0	1	1	1	1
	7	0	4	-4	(7)	35	2	30	14

Step: 3 :- $\frac{T^2}{N} = \frac{(7)^2}{12} = 4.083$

Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N} = 35 + 2 + 30 + 14 - 4$

$TSS = 76.917$

Step: 5 :- $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$
 $= \frac{(7)^2}{3} + 0 + \frac{(4)^2}{3} + \frac{(-4)^2}{3} - 4.083 \Rightarrow 22.917$
 $\Rightarrow \boxed{8.167 = SSC}$

Step: 6 :- $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} - \frac{T^2}{N} = \frac{49}{4} + 0 - 4.083$

$SSR = 8.167$

Step: 7 :- $SSE = TSS - SSC - SSR$
 $= 76.917 - 22.917 - 8.167$

$SSE = 45.833$

Step: 9 :- conclusion

	B/w Column	B/w Rows
T.V	4.76	19.23
C.V	1.00	1.87
Result	$T.V > C.V$	$T.V > C.V$
Conclusion	H_0 may be Accepted	H_0 may be Accepted

Latin Square Design [Three way classification]

Step: 1 :- N

Step: 2 :- T

Step: 3 :- $\frac{T^2}{N}$ Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \dots - \frac{T^2}{N}$ Step: 5 :- $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} \dots - \frac{T^2}{N}$ [N_1 - no of elements in each column]Step: 6 :- $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} \dots - \frac{T^2}{N}$ [N_2 - no. of elements in each row]

Step: 7 :- SSK [Rearrange the treatment & follow SSR]

Step: 8 :- $SSE = TSS - SSC - SSR - SSK$

Step: 9 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table value @ 1% (or) 5%
B/w Columns	SSC	C-1	$MSC = \frac{SSC}{C-1}$	$F_C = \frac{MSC}{MSE}$ (or) $\frac{MSE}{MSC}$	$F_C(\gamma_1, \gamma_2)$
B/w Rows	SSR	R-1	$MSR = \frac{SSR}{R-1}$	$F_R = \frac{MSR}{MSE}$ (or) $\frac{MSE}{MSR}$	$F_R(\gamma_1, \gamma_2)$
B/w treatment	SSK	K-1	$MSK = \frac{SSK}{K-1}$	$F_K = \frac{MSK}{MSE}$ (or) $\frac{MSE}{MSK}$	$F_K(\gamma_1, \gamma_2)$
Error	SSE	$\frac{(K-1)(K-2)}{(K-1)(K-2)}$	$MSE = \frac{SSE}{(K-1)(K-2)}$	$F > 1$	

Step: 10 :- Conclusion.

1. A Variable that was concluded on wheat with a Varieties in LSD. The plan of experiment per plot yield are given below.

C 25	B 23	A 20	D 20
A 19	D 19	C 21	B 18
B 1	A 14	D 17	C 20
D 17	C 20	B 21	A 15

Soln:

H_0 : There is no significant diff. b/w treatments.

H_1 : There is a significant diff. b/w treatments.

(X-20)

X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
5	3	0	0	8	25	9	0	0
-1	-1	1	-2	-3	1	1	1	4
-19	-6	-3	0	-28	361	36	9	0
-3	0	1	-5	-7	9	0	1	25
-18	-4	-1	-7	-30	396	46	11	29

Step: 1 :- $N = 16$

Step: 2 :- $T = -30$

Step: 3 :- $\frac{T^2}{N} = \frac{(-30)^2}{16} = 56.25$

Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$

$$= 396 + 46 + 11 + 29 - 56.25$$

$$TSS = 425.75$$

$$\text{Step: 5 :- } SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{(-18)^2}{4} + \frac{(-4)^2}{4} + \frac{(-1)^2}{4} + \frac{(-7)^2}{4} - 56.25$$

$$SSC = 41.25$$

$$\text{Step: 6 :- } SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$$

$$= \frac{(8)^2}{4} + \frac{(-3)^2}{4} + \frac{(-28)^2}{4} + \frac{(-7)^2}{4} - 56.25$$

$$SSR = 170.25$$

Step: 7 :- SSK

Rearrange the treatments

A	-1	-6	0	-5	$\Sigma A = -12$
B	-19	3	1	-2	$\Sigma B = -17$
C	5	0	1	0	$\Sigma C = 6$
D	-3	-1	-3	0	$\Sigma D = -7$

$$SSK = \frac{(-12)^2}{4} + \frac{(-17)^2}{4} + \frac{(6)^2}{4} + \frac{(-7)^2}{4} - 56.25$$

$$SSK = 73.25$$

Step: 8 :- SSE

$$= TSS - SSC - SSR - SSK$$

$$= 425.75 - 41.25 - 170.25 - 73.25$$

$$SSE = 141$$

Step: 9 :- ANOVA TABLE.

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value @ 5%
B/w Columns	SSC = 41.25	C-1 = 4-1 = 3	MSC = $\frac{SSC}{C-1}$ = $\frac{41.25}{3}$ = 13.75	$F_C = \frac{MSE}{MSC}$ = $\frac{23.5}{13.75}$ = 1.70	$F_C(6,3)$ = 8.94
B/w rows	SSR = 170.25	r-1 = 4-1 = 3	MSR = $\frac{SSR}{r-1}$ = $\frac{170.25}{3}$ = 56.75	$F_R = \frac{MSR}{MSE}$ = $\frac{56.75}{23.5}$ = 2.414	$F_C(3,6)$ = 4.76
B/w treatment	SSK = 73.25	K-1 = 3	MSK = $\frac{SSK}{3}$ = $\frac{73.25}{3}$ = 24.4	$F_K = \frac{MSK}{MSE}$ = $\frac{24.4}{23.5}$ = 1.03	$F_K(3,6)$ = 4.76
Error	SSE = 141	(K-1)(K-2) = 3(2) = 6	MSE = $\frac{SSE}{6}$ = $\frac{141}{6}$ = 23.5	$F > 1$	

Step: 10 :- conclusion

	B/w Column	B/w Row	B/w Treatment
T.V	8.94	4.76	4.76
C.V	1.70	2.414	1.03
Result	$T.V > C.V$	$T.V > C.V$	$T.V > C.V$
Conclusion	H_0 may be Accepted	H_0 may be Accepted	H_0 may be Accepted

2. The following is a LSD when 4 varieties of seeds are being tested. Perform an analysis and state the conclusion.

A 105	B 95	C 125	D 115
C 115	D 125	A 105	B 105
D 115	C 95	B 105	A 115
B 95	A 135	D 95	C 115

Soln:

H_0 : There is no significant difference b/w the variety of seeds.

H_1 : There is a significant difference b/w the variety of seeds.

$$\frac{X-95}{5}$$

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	2 _A	0 _B	6 _C	4 _D	12	4	0	36	16
Y_2	4 _A	6 _B	2 _A	2 _B	14	16	36	4	4
Y_3	4 _A	0 _C	2 _B	4 _A	10	16	0	4	16
Y_4	0 _A	8 _A	0 _C	4 _C	12	0	64	0	16
	10	14	0	14	(48)	36	100	44	52

Step: 1 :- $N = 16$

Step: 2 :- $T = 48$

Step: 3 :- $\frac{T^2}{N} = 144$

Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 36 + 100 + 44 + 52 - 144$

$TSS = 88$

$$\begin{aligned}\text{Step : 5 :- } SSC &= \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N} \\ &= \frac{(10)^2}{4} + \frac{(14)^2}{4} + \frac{(10)^2}{4} + \frac{(14)^2}{4} - 144\end{aligned}$$

$$SSC = 4$$

$$\begin{aligned}\text{Step : 6 :- } SSR &= \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N} \\ &= \frac{(12)^2}{4} + \frac{(14)^2}{4} + \frac{(10)^2}{4} + \frac{(12)^2}{4} - 144\end{aligned}$$

$$SSR = 2$$

Step : 7 :- SSK

Re-arrange the table.

$$A \quad 2 \quad 8 \quad 2 \quad 4 \quad \Sigma A = 16$$

$$B \quad 0 \quad 0 \quad 2 \quad 2 \quad \Sigma B = 4$$

$$C \quad 4 \quad 0 \quad 6 \quad 4 \quad \Sigma C = 14$$

$$D \quad 4 \quad 6 \quad 0 \quad 4 \quad \Sigma D = 14$$

$$\begin{aligned}SSK &= \frac{(16)^2}{4} + \frac{(4)^2}{4} + \frac{(14)^2}{4} + \frac{(14)^2}{4} - 144 \\ &= \frac{256}{4} + \frac{16}{4} + \frac{196}{4} + \frac{196}{4} - 144\end{aligned}$$

$$SSK = 22$$

Step : 8 :- SSE

$$\begin{aligned}&= TSS - SSC - SSR - SSK \\ &= 88 - 4 - 2 - 22\end{aligned}$$

$$SSE = 60$$

Step: 9 :- ANOVA TABLE

Source of Variance	Sum of Square	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table value at 5%
B/w Columns	SSC = 4	$C-1$ = $4-1$ = 3	$MSC = \frac{SSC}{C-1}$ = $\frac{4}{3} = 1.33$	$F_C = \frac{MSE}{MSC}$ = $\frac{10}{1.333} = 7.5$	$F_2(3, 12)$ = (6.3) = 8.94
B/w Rows	SSR = 2	$r-1$ = 3	$MSR = \frac{SSR}{r-1}$ = $\frac{2}{3} = 0.6666$	$F_R = \frac{MSE}{MSR}$ = $\frac{10}{0.6666}$ = 15	$F_R(6, 3)$ = 8.94
B/w treatments	SSK = 22	$K-1$ = 3	$MSK = \frac{SSK}{K-1}$ = $\frac{22}{3} = 7.3333$	$F_K = \frac{MSE}{MSK}$ = $\frac{10}{7.3333}$ $F_K = 1.36$	$F_K(6, 3)$ = 8.94
Error	SSE = 60	$(k-1)$ $(15-2)$ = 6	$MSE = \frac{SSE}{6}$ = $\frac{60}{6} = 10$	$F > 1$	

Step: 10 :- Conclusion

	B/w Columns	B/w Rows	B/w Treatments
T.V	8.94	8.94	8.94
C.V	7.15	15	1.36
Result	$T.V > C.V$	$T.V < C.V$	$T.V > C.V$
Conclusion	H_0 may be Accepted	H_0 is Rejected	H_0 may be Accepted

3. The following data resulted from an experiment to compare 3 burners A, B, C. A LSD was used as the test were made on 3 engines and were spread over 3 days.

A 16 B 17 C 20
 B 16 C 21 A 15
 C 15 A 12 B 13

Soln :-

H_0 : There is no significant difference b/w treatments

H_1 : There is a significant difference b/w treatments.

	X_1	X_2	X_3	Total	X_1^2	X_2^2	X_3^2
Y_1	6	7	10	23	36	49	100
Y_2	6	11	5	22	36	121	25
Y_3	5	2	3	10	25	4	9
	17	20	18	(55)	97	174	134

Step: 1 :- $N = 9$

Step: 2 :- $T = 55$

Step: 3 :- $\frac{T^2}{N} = 336.11$

Step: 4 :- TSS

$$= \sum X_1^2 + \sum X_2^2 + \sum X_3^2 - \frac{T^2}{N}$$

$$= 97 + 174 + 134 - 336.11$$

$$\boxed{TSS = 68.89}$$

Step: 5 :- SSC

$$= \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} - \frac{T^2}{N}$$

$$= \frac{(17)^2}{3} + \frac{(20)^2}{3} + \frac{(18)^2}{3} - 336.11$$

$$SSC = 1.55$$

Step: 6 :- SSR

$$= \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} - \frac{T^2}{N}$$

$$= \frac{(23)^2}{3} + \frac{(22)^2}{3} + \frac{(10)^2}{3} - 336.11 \Rightarrow SSR = 34.87$$

Step: 7 :- SSK

Re-arranging the table

A	6	2	5	13
B	6	7	3	16
C	5	11	10	26

$$SSK = \frac{(13)^2}{3} + \frac{(16)^2}{3} + \frac{(26)^2}{3} - 336.11$$

$$SSK = 30.89$$

Step: 8 :- SSE = TSS - SSC - SSR - SSK

$$= 68.89 - 1.55 - 34.87 - 30.89 \Rightarrow SSE = 1.56$$

Step: 9 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table value @ 5%
B/w Columns	SSC = 1.55	C-1 = 3-1 = 2	MSC = $\frac{SSC}{C-1}$ = $\frac{1.55}{2} = 0.775$	$F_C = \frac{MSE}{MSC}$ = $\frac{0.78}{0.775} = 1.01$	$F_C(2,2)$ = 19.00
B/w Rows	SSR = 34.87	r-1 = 2	MSR = $\frac{SSR}{r-1}$ = $\frac{34.87}{2} = 17.445$	$F_R = \frac{MSR}{MSE}$ = $\frac{17.445}{0.78} = 22.36$	$F_R(2,2)$ = 19.00
B/w treatment	SSK = 30.89	K-1 = 2	MSK = $\frac{SSK}{2}$ = $\frac{30.89}{2} = 15.445$	$F_K = \frac{MSK}{MSE}$ = $\frac{15.445}{0.78} = 19.79$	$F_K(2,2)$ = 19.00
Error	SSE = 1.56	(k-1)(k-2) = 2	MSE = $\frac{1.56}{2}$ = 0.78	$F > 1$	

Step 10 - Conclusion

	B/w Columns	B/w Rows	B/w Treatments
TV	19.00	19.00	19.00
CV	1.01	22.36	19.79
Result	$T.V > C.V$	$T.V < C.V$	$T.V < C.V$
Conclusion	H_0 may be Accepted	H_0 is Rejected	H_0 is Rejected

4. A farmer wishes to test the effects of 4 fertilizers A, B, C, D in the yield of wheat. The fertilizer are used in LSD and the results are tabulated. Perform an analysis of Variance.

A 18	C 21	D 25	B 11
D 22	B 12	A 15	C 19
B 15	A 20	C 23	D 24
C 22	D 21	B 10	A 17

Soln:

- H_0 : There is no significant diff. b/w the fertilizers
 H_1 : There is a significant diff. b/w the fertilizers

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	3	6	10	4	15	9	36	100	16
Y_2	7	-3	0	4	8	49	9	0	16
Y_3	0	5	8	9	22	0	25	64	81
Y_4	7	6	-5	2	10	49	36	25	4
	17	14	13	11	55	107	106	189	117

Step: 1 :- $N = 16$

Step: 2 :- $T = 55$

Step: 3 :- $\frac{T^2}{N} = 189.06$

Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 107 + 106 + 189 + 117 - 189.06$

$TSS = 329.94$

Step: 5 :- $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$
 $= \frac{(17)^2}{4} + \frac{(14)^2}{4} + \frac{(13)^2}{4} + \frac{(11)^2}{4} - 189.06$

$SSC = 4.69$

Step: 6 :- $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$
 $= \frac{(15)^2}{4} + \frac{(8)^2}{4} + \frac{(22)^2}{4} + \frac{(10)^2}{4} - 189.06$

$SSR = 29.19$

Step: 7 :- SSK

					Total
A	3	5	0	2	10
B	0	-3	-5	-4	-12
C	7	6	8	4	25
D	7	6	10	9	32

$$SSK = \frac{(10)^2}{4} + \frac{(-12)^2}{4} + \frac{(25)^2}{4} + \frac{(32)^2}{4} - 189.06$$

$\Rightarrow SSK = 284.19$

Step: 8 :- SSE

$$= TSS - SSC - SSR - SSK$$

$$= 329.94 - 4.69 - 29.19 - 284.19$$

$$SSE = 11.87$$

Step: 9 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table Value @ 5%
B/w Columns	SSC = 4.69	c-1 = 4-1 = 3	MSC = $\frac{SSC}{c-1}$ = $\frac{4.69}{3} = 1.56$	$F_c = \frac{MSE}{MSC} = \frac{1.97}{1.56}$ = 1.26	$F_c(6,3)$ = 5.94
B/w Rows	SSR = 29.19	r-1 = 4-1 = 3	MSR = $\frac{SSR}{r-1}$ = $\frac{29.19}{3} = 9.73$	$F_R = \frac{MSR}{MSE} = \frac{9.73}{1.97}$ = 4.93	$F_R(3,6)$ = 4.76
B/w treatment	SSK = 284.19	K-1 = 4-1 = 3	MSK = $\frac{SSK}{K-1}$ = $\frac{284.19}{3} = 94.73$	$F_k = \frac{MSK}{MSE} = \frac{94.73}{1.97}$ = 48.08	$F_k(3,6)$ = 4.76
Error	SSE = 11.87	(k-1)(k-2) = 3x2 = 6	MSE = $\frac{SSE}{6}$ = $\frac{11.87}{6} = 1.97$	$F > 1$	

Step: 10 :- Conclusion

	B/w Columns	B/w Rows	B/w Treatments
T.V	5.94	4.76	4.76
C.V	1.26	4.93	48.08
Result	$T.V > C.V$	$T.V < C.V$	$T.V < C.V$
Conclusion	H_0 may be Accepted	H_0 is Rejected	H_0 is Rejected

5. Analyse the following.

S 122	P 121	R 123	Q 122
Q 124	R 123	P 122	S 125
P 120	Q 119	S 120	R 121
R 122	S 123	Q 121	P 122

Soln:

H_0 : There is no significant difference b/w treatments.

H_1 : There is a significant difference b/w treatments.

(X-120)

	X_1	X_2	X_3	X_4	Total	X_1^2	X_2^2	X_3^2	X_4^2
Y_1	2	1	3	2	8	4	1	9	4
Y_2	4	3	2	5	14	16	9	4	25
Y_3	0	-1	0	1	0	0	1	0	1
Y_4	2	3	1	2	8	4	9	1	4
	8	6	6	10	(30)	24	20	14	34

Step: 1 :- $N = 16$

Step: 2 :- $T = 30$

Step: 3 :- $\frac{T^2}{N} = 56.25$

Step: 4 :- $TSS = \sum X_1^2 + \sum X_2^2 + \sum X_3^2 + \sum X_4^2 - \frac{T^2}{N}$
 $= 24 + 20 + 14 + 34 - 56.25 \Rightarrow TSS = 35.75$

Step: 5 :- $SSC = \frac{(\sum X_1)^2}{N_1} + \frac{(\sum X_2)^2}{N_1} + \frac{(\sum X_3)^2}{N_1} + \frac{(\sum X_4)^2}{N_1} - \frac{T^2}{N}$
 $= \frac{64}{4} + \frac{36}{4} + \frac{36}{4} + \frac{100}{4} - 56.25 \Rightarrow SSC = 2.75$

Step: 6 :- $SSR = \frac{(\sum Y_1)^2}{N_2} + \frac{(\sum Y_2)^2}{N_2} + \frac{(\sum Y_3)^2}{N_2} + \frac{(\sum Y_4)^2}{N_2} - \frac{T^2}{N}$
 $= \frac{64}{4} + \frac{196}{4} + 0 + \frac{64}{4} - 56.25 \Rightarrow SSR = 24.75$

Step: 7 :- SSK

P	0	1	2	2	5
Q	4	-1	1	2	6
R	2	3	3	1	9
S	2	3	0	5	10

$$SSK = \frac{25}{4} + \frac{36}{4} + \frac{81}{4} + \frac{100}{4} - 56.25$$

$$SSK = 4.25$$

Step: 8 :- $SSE = TSS - SSC - SSR - SSK \Rightarrow SSE = 4$

Step: 9 :- ANOVA TABLE

Source of Variance	Sum of Squares	Degrees of Freedom	Mean Sum of Squares	Variance Ratio	Table value @ 5%
B/w Columns	SSC = 2.75	$C-1$ = 3	$MSC = \frac{SSC}{C-1}$ = $\frac{2.75}{3} = 0.9166$	$F_C = \frac{MSC}{MSE}$ = $\frac{0.9166}{0.6666} = 1.37$	$F_C(3,6)$ = 8.44 = 4.76
B/w Rows	SSR = 24.75	$r-1$ = 3	$MSR = \frac{SSR}{r-1}$ = $\frac{24.75}{3} = 8.25$	$F_R = \frac{8.25}{0.6666}$ = 12.38	$F_R(3,6)$ = 4.76
B/w Treatments	SSK = 4.25	$k-1$ = 3	$MSK = \frac{SSK}{k-1}$ = $\frac{4.25}{3} = 1.41$	$F_K = \frac{MSK}{MSE} = \frac{1.41}{0.666}$ = 2.11	$F_K(3,6)$ = 4.76
Error	SSE = 4	$(k-1)(k-2)$ = 6	$MSE = \frac{SSE}{6}$ = $\frac{4}{6} = 0.6666$	$F > 1$	

Step: 10 :- Conclusion

	B/w Columns	B/w Rows	B/w Treatments
T.V	4.76	4.76	4.76
C.V	1.37	12.38	2.11
Result	$T.V > C.V$	$T.V < C.V$	$T.V > C.V$
Conclusion	H_0 may be Accepted	H_0 is Rejected	H_0 may be Accepted